DATA 5321 - Group Final Project

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```r
# Read Data
bike_raw <- read.csv("/Users/dcnguyen060899/Downloads/SU_Winter_2026/DATA_5321/DATA5321_Bike_Share_Daily_Dat

# Re-assign values of variables
bike_raw <- bike_raw %>%
  mutate(
    season     = factor(season,     levels = 1:4,  labels = c("Spring", "Summer", "Fall", "Winter")),
    mnth       = factor(mnth,       levels = 1:12),
    holiday    = factor(holiday,    levels = c(0,1), labels = c("No", "Yes")),
    weekday    = factor(weekday,    levels = 0:6,  labels = c("Sun","Mon","Tue","Wed","Thu","Fri","Sat")),
    workingday = factor(workingday, levels = c(0,1), labels = c("No", "Yes")),
    weathersit = factor(weathersit, levels = 1:4,  labels = c("Clear", "Mist/Cloudy", "Light Rain/Snow", "He
  )

# Filter for 2011
bike_2011 <- bike_raw %>% filter(yr == 0)
```

```r
# let's first understand the data dimension, variable type, any missing value, its basic summary statistic
dim(bike_2011)
```

```
## [1] 365  16
```

```r
str(bike_2011)
```

```
## 'data.frame':    365 obs. of  16 variables:
##  $ instant   : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ dteday    : chr  "2011-01-01" "2011-01-02" "2011-01-03" "2011-01-04" ...
##  $ season    : Factor w/ 4 levels "Spring","Summer",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ yr        : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ mnth      : Factor w/ 12 levels "1","2","3","4",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ holiday   : Factor w/ 2 levels "No","Yes": 1 1 1 1 1 1 1 1 1 1 ...
##  $ weekday   : Factor w/ 7 levels "Sun","Mon","Tue",..: 7 1 2 3 4 5 6 7 1 2 ...
##  $ workingday: Factor w/ 2 levels "No","Yes": 1 1 2 2 2 2 2 1 1 2 ...
##  $ weathersit: Factor w/ 4 levels "Clear","Mist/Cloudy",..: 2 2 1 1 1 1 2 2 1 1 ...
##  $ temp      : num  0.344 0.363 0.196 0.2 0.227 ...
##  $ atemp     : num  0.364 0.354 0.189 0.212 0.229 ...
##  $ hum       : num  0.806 0.696 0.437 0.59 0.437 ...
##  $ windspeed : num  0.16 0.249 0.248 0.16 0.187 ...
##  $ casual    : int  331 131 120 108 82 88 148 68 54 41 ...
##  $ registered: int  654 670 1229 1454 1518 1518 1362 891 768 1280 ...
##  $ cnt       : int  985 801 1349 1562 1600 1606 1510 959 822 1321 ...
```

```r
summary(bike_2011)
```

```
##     instant       dteday             season         yr         mnth
##  Min.   :  1   Length:365         Spring:90   Min.   :0   1      : 31
##  1st Qu.: 92   Class :character   Summer:92   1st Qu.:0   3      : 31
##  Median :183   Mode  :character   Fall  :94   Median :0   5      : 31
```

```
##  Mean   :183                      Winter:89   Mean   :0   7      : 31
##  3rd Qu.:274                                  3rd Qu.:0   8      : 31
##  Max.   :365                                  Max.   :0   10     : 31
##                                                           (Other):179
##  holiday   weekday  workingday           weathersit       temp
##  No :355   Sun:52   No :115    Clear          :226   Min.   :0.05913
##  Yes: 10   Mon:52   Yes:250    Mist/Cloudy    :124   1st Qu.:0.32500
##            Tue:52              Light Rain/Snow: 15   Median :0.47917
##            Wed:52              Heavy Rain/Snow:  0   Mean   :0.48666
##            Thu:52                                    3rd Qu.:0.65667
##            Fri:52                                    Max.   :0.84917
##            Sat:53
##      atemp              hum           windspeed           casual
##  Min.   :0.07907   Min.   :0.0000   Min.   :0.02239   Min.   :   9.0
##  1st Qu.:0.32195   1st Qu.:0.5383   1st Qu.:0.13558   1st Qu.: 222.0
##  Median :0.47285   Median :0.6475   Median :0.18690   Median : 614.0
##  Mean   :0.46684   Mean   :0.6437   Mean   :0.19140   Mean   : 677.4
##  3rd Qu.:0.61238   3rd Qu.:0.7421   3rd Qu.:0.23508   3rd Qu.: 871.0
##  Max.   :0.84090   Max.   :0.9725   Max.   :0.50746   Max.   :3065.0
##
##    registered        cnt
##  Min.   : 416   Min.   : 431
##  1st Qu.:1730   1st Qu.:2132
##  Median :2915   Median :3740
##  Mean   :2728   Mean   :3406
##  3rd Qu.:3632   3rd Qu.:4586
##  Max.   :4614   Max.   :6043
##
```

```r
colSums(is.na(bike_2011))
```

```
##    instant     dteday     season         yr       mnth    holiday    weekday
##          0          0          0          0          0          0          0
## workingday weathersit       temp      atemp        hum  windspeed     casual
##          0          0          0          0          0          0          0
## registered        cnt
##          0          0
```

- There are 365 rows, so that is basically 365 days in a year in 2011, and 16 columns
- Factors are correctly assigned to categorical feature like season, mnth, holiday, weekday, workingday, weathersit
- our outcome variable are cnt, casual, registed and they are integers
- instant is like ID indexing and yr encode as year like 0 for 20113, 1 for 2012, so these we probably won't include as our predictors.

For summary: - 90 days each for each seasons so data is balanced - weathersit:mostly clear with 226, Mist/Claudy (124), very few Light Rain (15), zero Heavy Rain

Good news, no missing data.

However, considering three things: 1. There is no observation heavy rain or snow. Might bias in modelling. 2. hum minimum is 0 seems unsual 3. cnt has a wide range, good to check its distribution in EDA

Now we have a set of continuous and categorical variables.

```r
# Continuous variables
bike_2011 %>%
  select(temp, atemp, hum, windspeed, cnt) %>%
  summary()
```

```
##       temp             atemp              hum           windspeed
##  Min.   :0.05913   Min.   :0.07907   Min.   :0.0000   Min.   :0.02239
##  1st Qu.:0.32500   1st Qu.:0.32195   1st Qu.:0.5383   1st Qu.:0.13558
```

```
##  Median :0.47917   Median :0.47285   Median :0.6475   Median :0.18690
##  Mean   :0.48666   Mean   :0.46684   Mean   :0.6437   Mean   :0.19140
##  3rd Qu.:0.65667   3rd Qu.:0.61238   3rd Qu.:0.7421   3rd Qu.:0.23508
##  Max.   :0.84917   Max.   :0.84090   Max.   :0.9725   Max.   :0.50746
##       cnt
##  Min.   : 431
##  1st Qu.:2132
##  Median :3740
##  Mean   :3406
##  3rd Qu.:4586
##  Max.   :6043
```

```r
# Standard deviations
bike_2011 %>%
  select(temp, atemp, hum, windspeed, cnt) %>%
  summarise(across(everything(), sd))
```

```
##        temp     atemp      hum  windspeed      cnt
## 1 0.1895956 0.1688356 0.148744 0.07689011 1378.754
```

```r
# Categorical variables - counts and proportions
cat_vars <- c("season", "mnth", "holiday", "weekday", "workingday", "weathersit")

for(var in cat_vars) {
  cat("\n---", var, "---\n")
  tbl <- table(bike_2011[[var]])
  print(tbl)
  print(round(prop.table(tbl), 3))
}
```

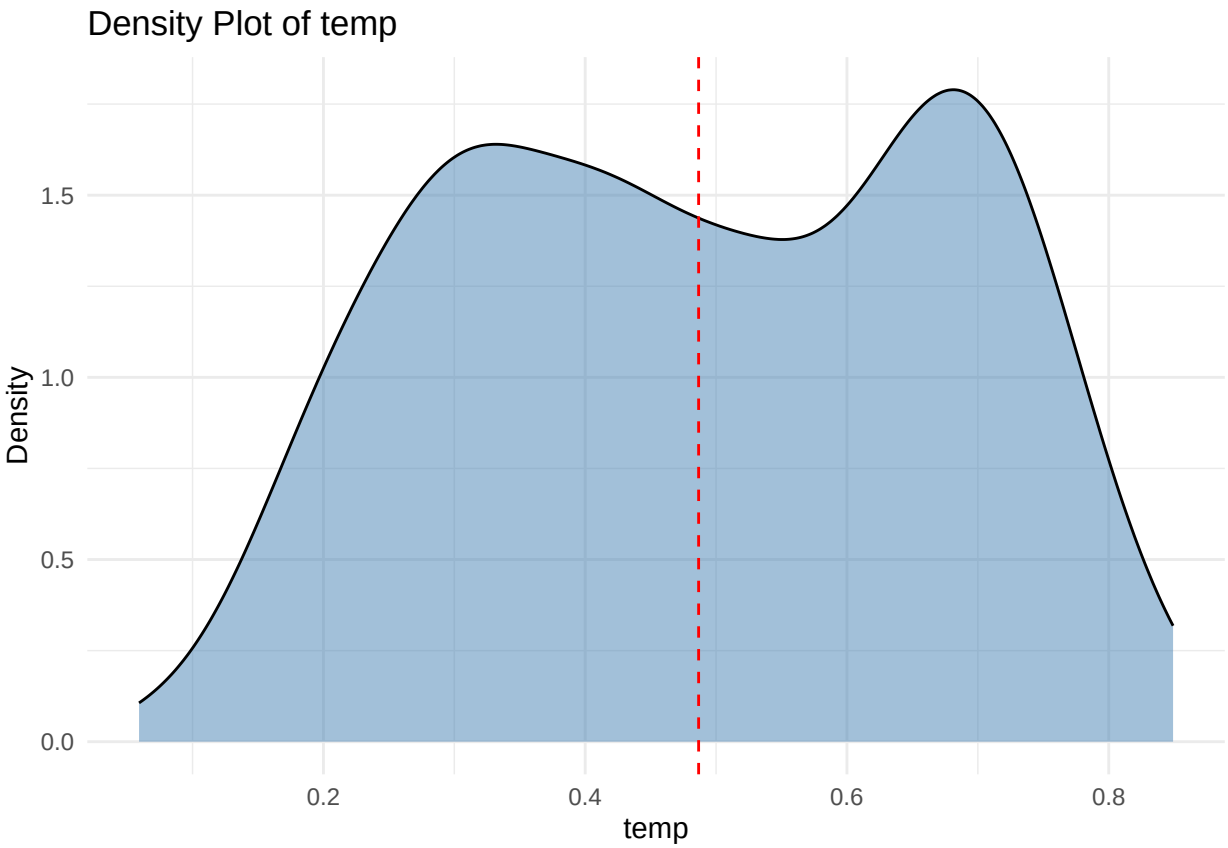
```
##
## --- season ---
##
## Spring Summer   Fall Winter
##     90     92     94     89
##
## Spring Summer   Fall Winter
##  0.247  0.252  0.258  0.244
##
## --- mnth ---
##
##  1  2  3  4  5  6  7  8  9 10 11 12
## 31 28 31 30 31 30 31 31 30 31 30 31
##
##     1     2     3     4     5     6     7     8     9    10    11    12
## 0.085 0.077 0.085 0.082 0.085 0.082 0.085 0.085 0.082 0.085 0.082 0.085
##
## --- holiday ---
##
##  No Yes
## 355  10
##
##    No   Yes
## 0.973 0.027
##
## --- weekday ---
##
## Sun Mon Tue Wed Thu Fri Sat
##  52  52  52  52  52  52  53
```

```
##
##   Sun   Mon   Tue   Wed   Thu   Fri   Sat
## 0.142 0.142 0.142 0.142 0.142 0.142 0.145
##
## --- workingday ---
##
##  No Yes
## 115 250
##
##    No   Yes
## 0.315 0.685
##
## --- weathersit ---
##
##           Clear     Mist/Cloudy Light Rain/Snow Heavy Rain/Snow
##             226             124              15               0
##
##           Clear     Mist/Cloudy Light Rain/Snow Heavy Rain/Snow
##           0.619           0.340           0.041           0.000
```

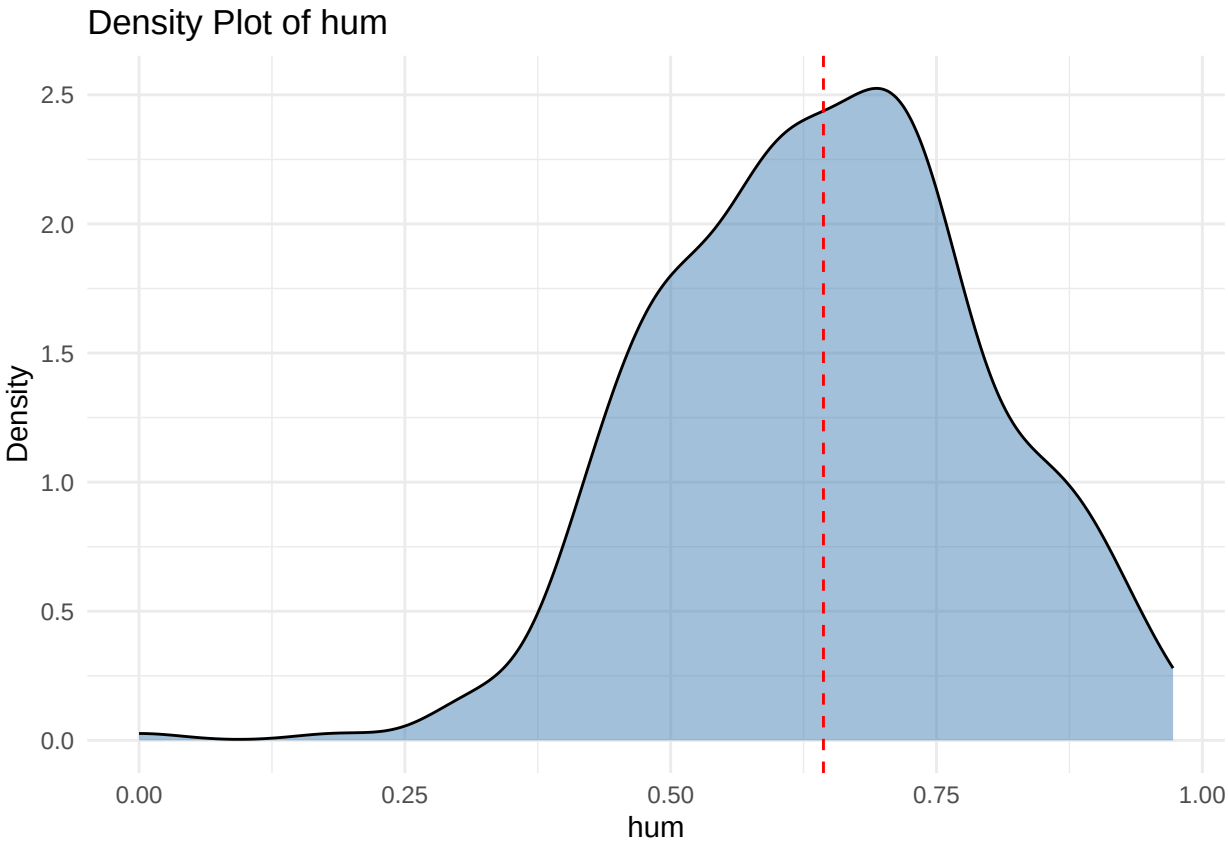
- Important thing to flag:

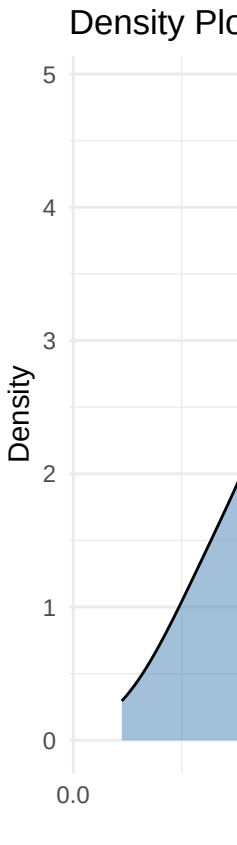
1. again hum min is 0, this could be possible outlier
2. again heavy rain or snow have no observation
3. holiday is very imbalanced
4. then also temp and atemp is likely multicollinear (this would violate the "no multicollinearity" (full-rank X) assumption of linear regression)

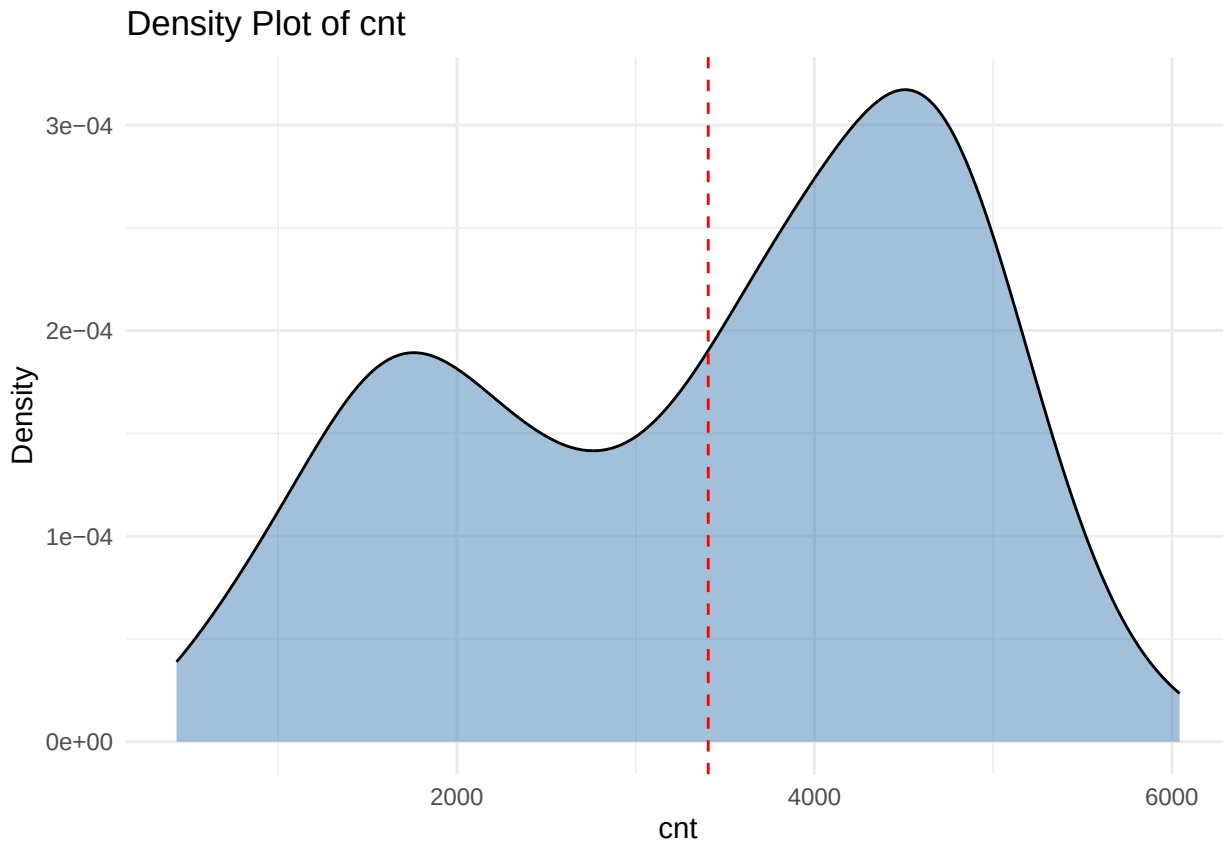
```r
cont_vars <- c("temp", "atemp", "hum", "windspeed", "cnt")

for(var in cont_vars) {
  p <- ggplot(bike_2011, aes(x = .data[[var]])) +
    geom_density(fill = "steelblue", alpha = 0.5) +
    geom_vline(aes(xintercept = mean(.data[[var]])),
               color = "red", linetype = "dashed") +
    labs(title = paste("Density Plot of", var),
         x = var, y = "Density") +
    theme_minimal()
  print(p)
}
```


Density Plot of temp
Density
1.5
1.0
0.5
0.0
0.2
0.4
0.6
0.8
temp

Density P
Density
2.0
1.5
1.0
0.5
0.0

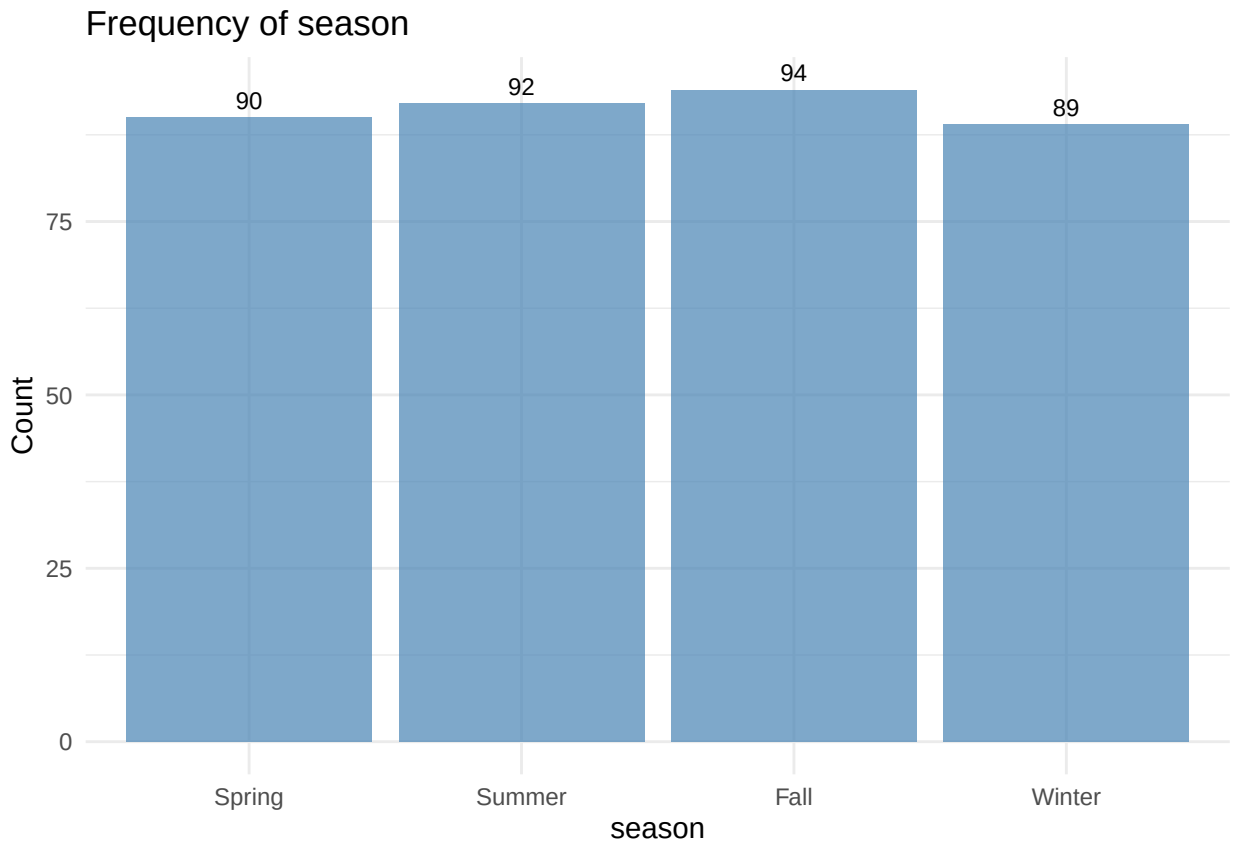

Density Plot of hum
Density
2.5
2.0
1.5
1.0
0.5
0.0
0.00
0.25
0.50
0.75
1.00
hum

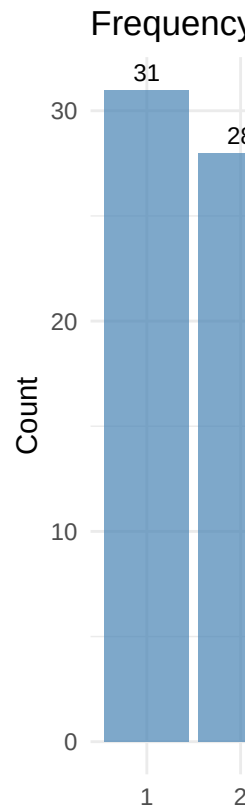

Density Plo
Density
5
4
3
2
1
0
0.0

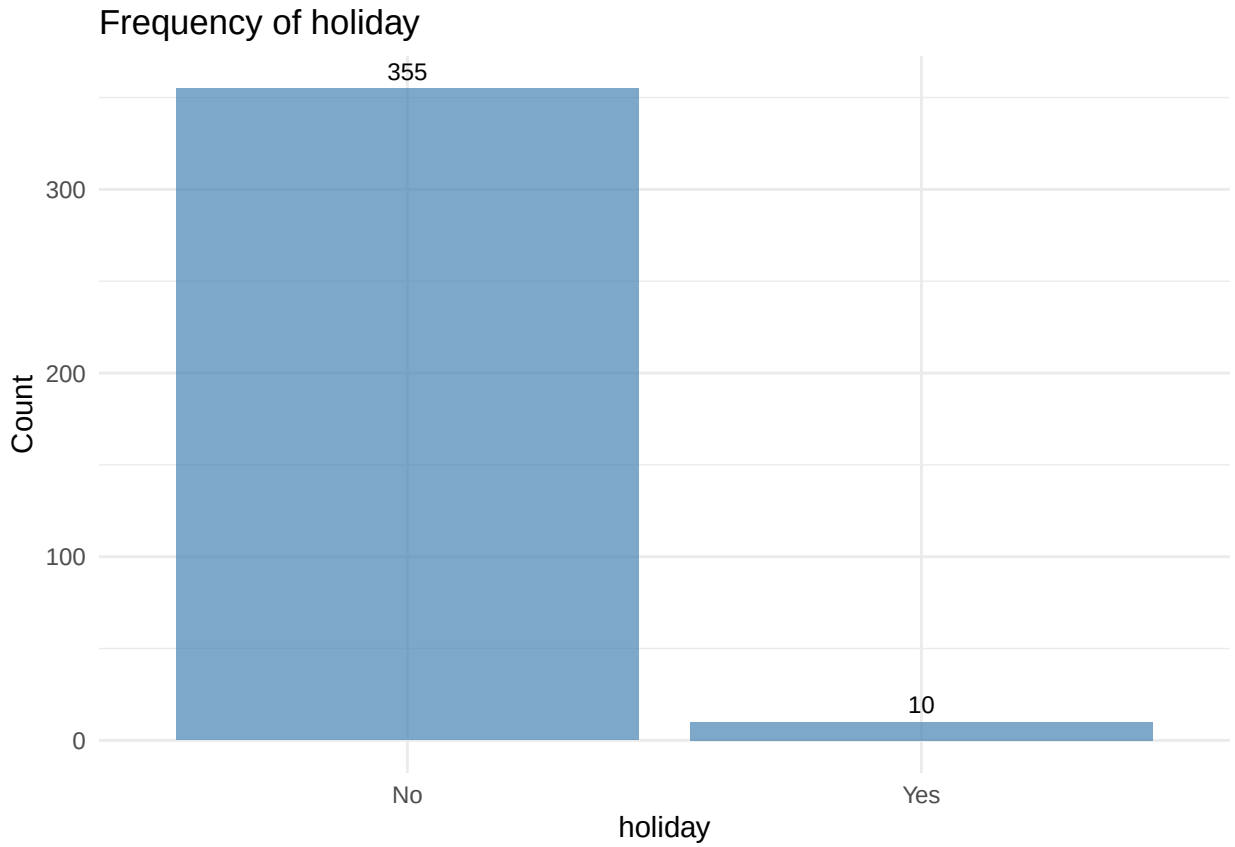


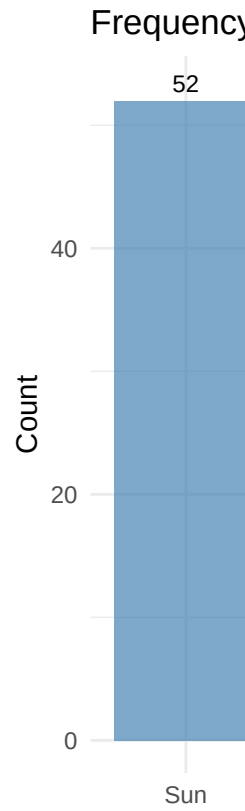
- temp: two peak, indicating bimodal
- atemp: also bimodal, indicating it correlate with tempt
- hum: left skewed, though look like a small bum at 0 even though stat said otherwise
- windspeed: right skewed
- cnt: also look bimodal, similar mirros the temp because intuitively, code days are low rental and warm days high rentals.

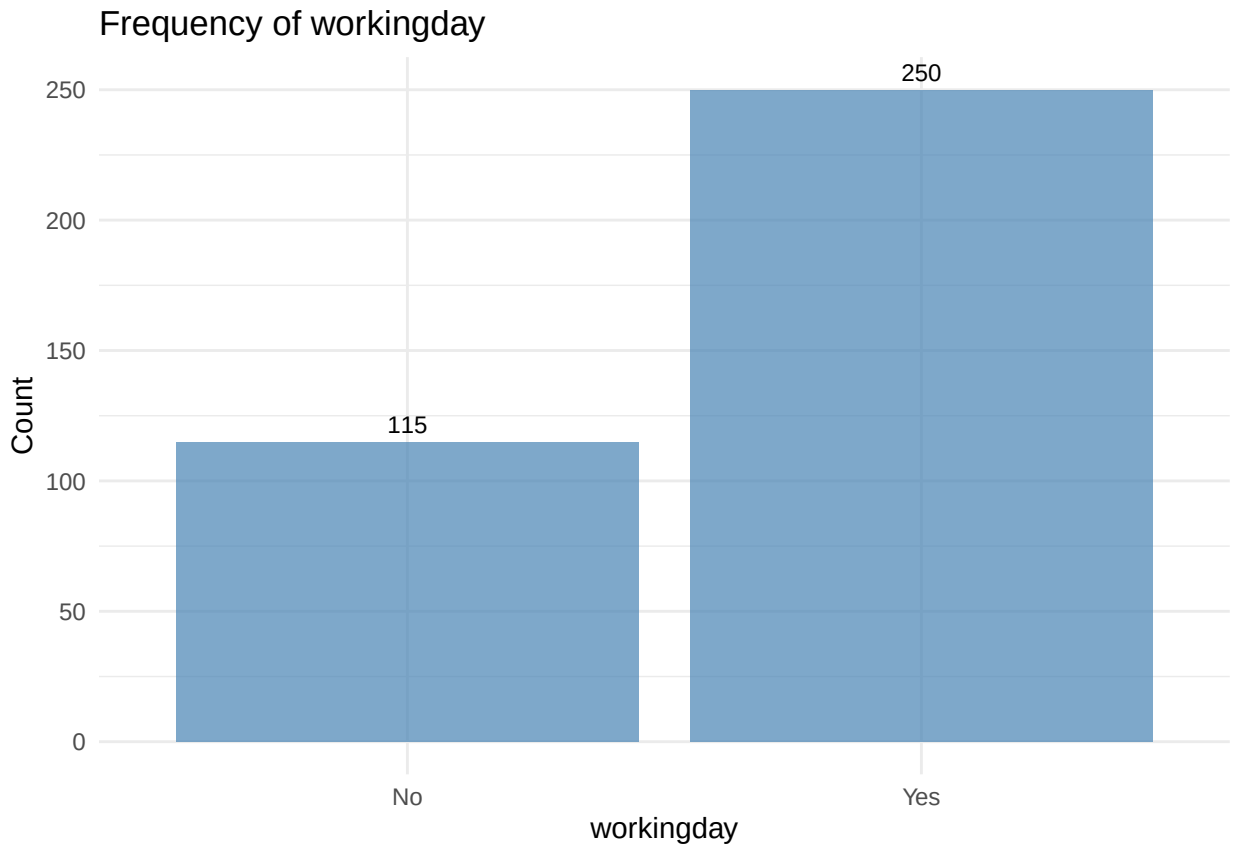
```
for(var in cat_vars) {
  p <- ggplot(bike_2011, aes(x = .data[[var]])) +
    geom_bar(fill = "steelblue", alpha = 0.7) +
    geom_text(stat = "count",
              aes(label = after_stat(count)),
              vjust = -0.5, size = 3) +
    labs(title = paste("Frequency of", var),
         x = var, y = "Count") +
    theme_minimal()
  print(p)
}
```

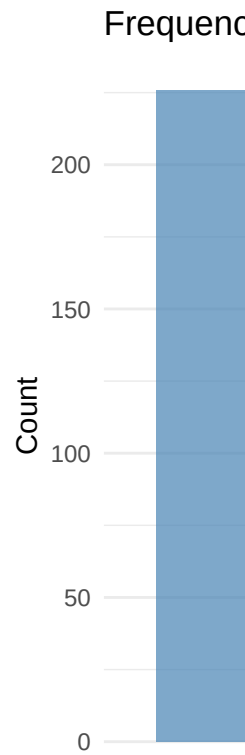

Frequency of season
90
92
94
89
75
50
25
0
Count
Spring
Summer
Fall
Winter
season


Frequency
31
30
20
10
0
Count
1


Frequency of holiday
355
10
300
200
100
0
Count
No
Yes
holiday


Frequency
52
40
20
0
Count
Sun





- season: near perfectly balanced
- mnth: also seem balanced too
- holiday: severely imbalanced
- weekday: perfectly balanced
- workingday: moderately balanced. Though note that workinday and weekday + holiday are logically redundant because working day is determined by weekday and holiday
- Since again, we repeated ourself that heavy rain/snow completely absent so let's drop it for modelling

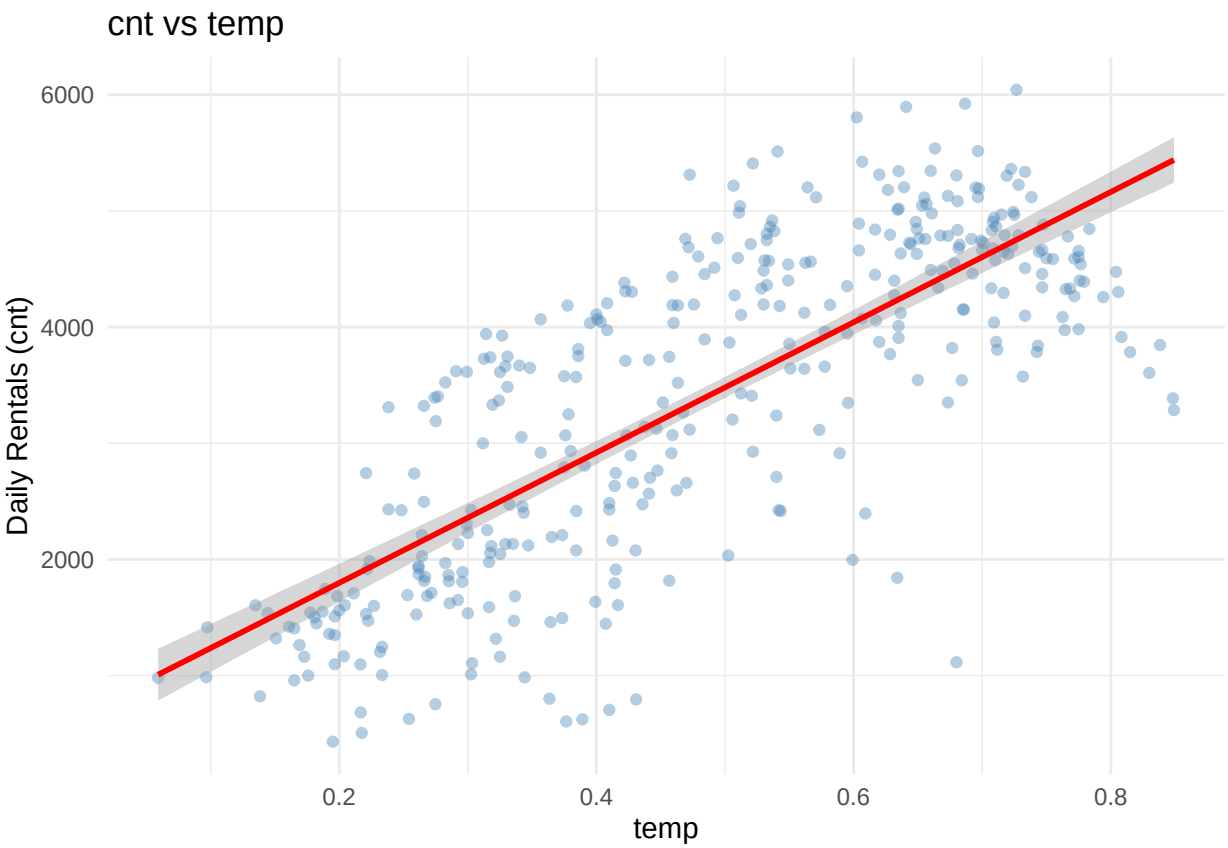
Key flag to watch out: - holiday imbalance - unreliable predictor - workingday is multicollinearity with weekday + holiday

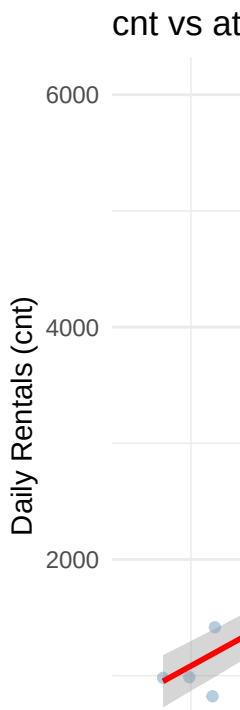
```
# dropping heavy rain/snow
bike_2011 <- bike_2011 %>%
  mutate(weathersit = droplevels(weathersit))

levels(bike_2011$weathersit)
```

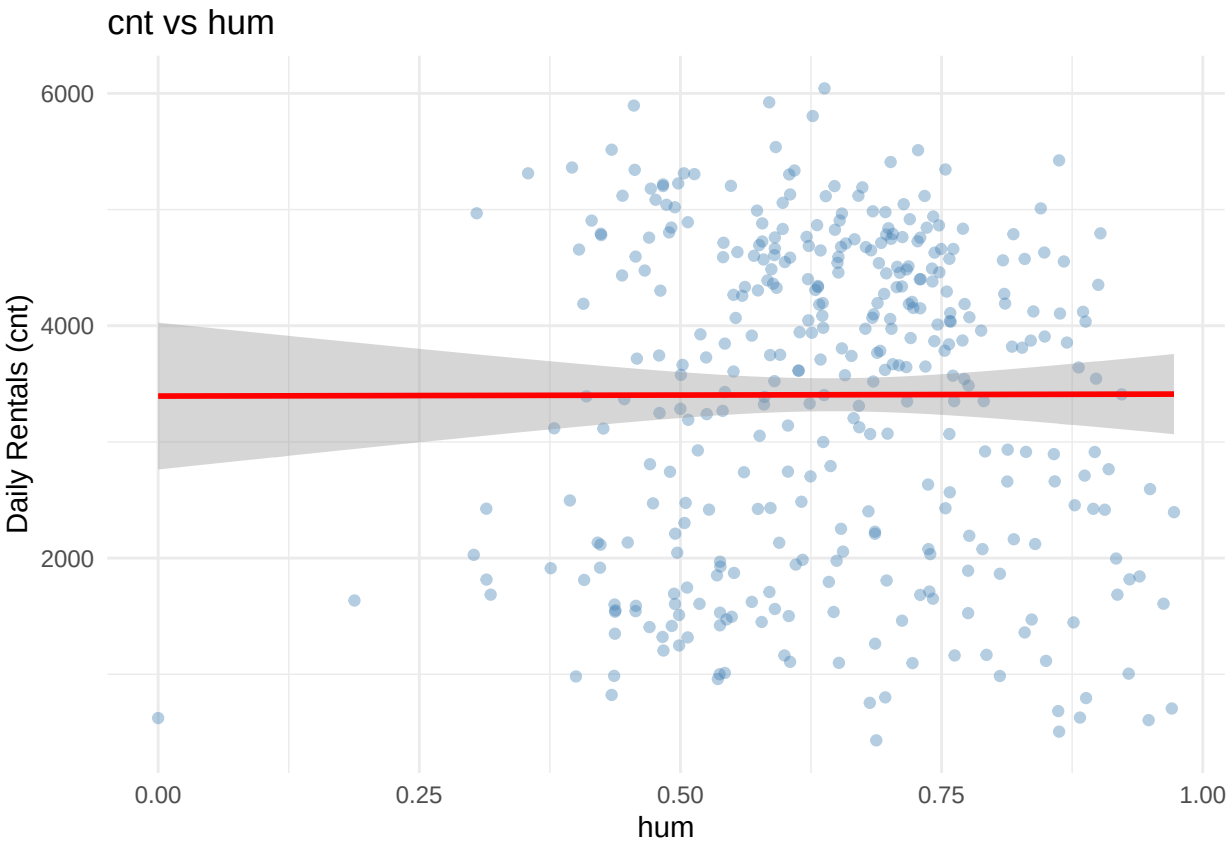
```
## [1] "Clear"           "Mist/Cloudy"     "Light Rain/Snow"
```

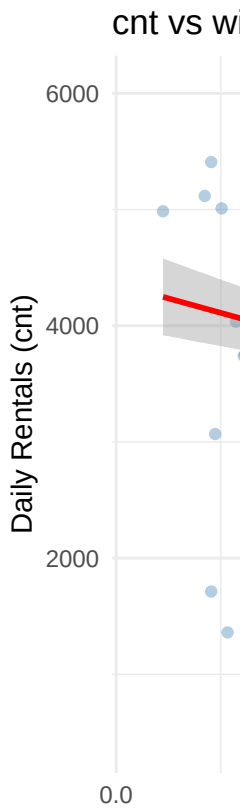
Now let's understand the relationship between predictors and cnt?

```
cont_preds <- c("temp", "atemp", "hum", "windspeed")

for(var in cont_preds) {
  p <- ggplot(bike_2011, aes(x = .data[[var]], y = cnt)) +
    geom_point(alpha = 0.4, color = "steelblue") +
    geom_smooth(method = "lm", color = "red", se = TRUE) +
    labs(title = paste("cnt vs", var),
         x = var, y = "Daily Rentals (cnt)") +
    theme_minimal()
  print(p)
}
```





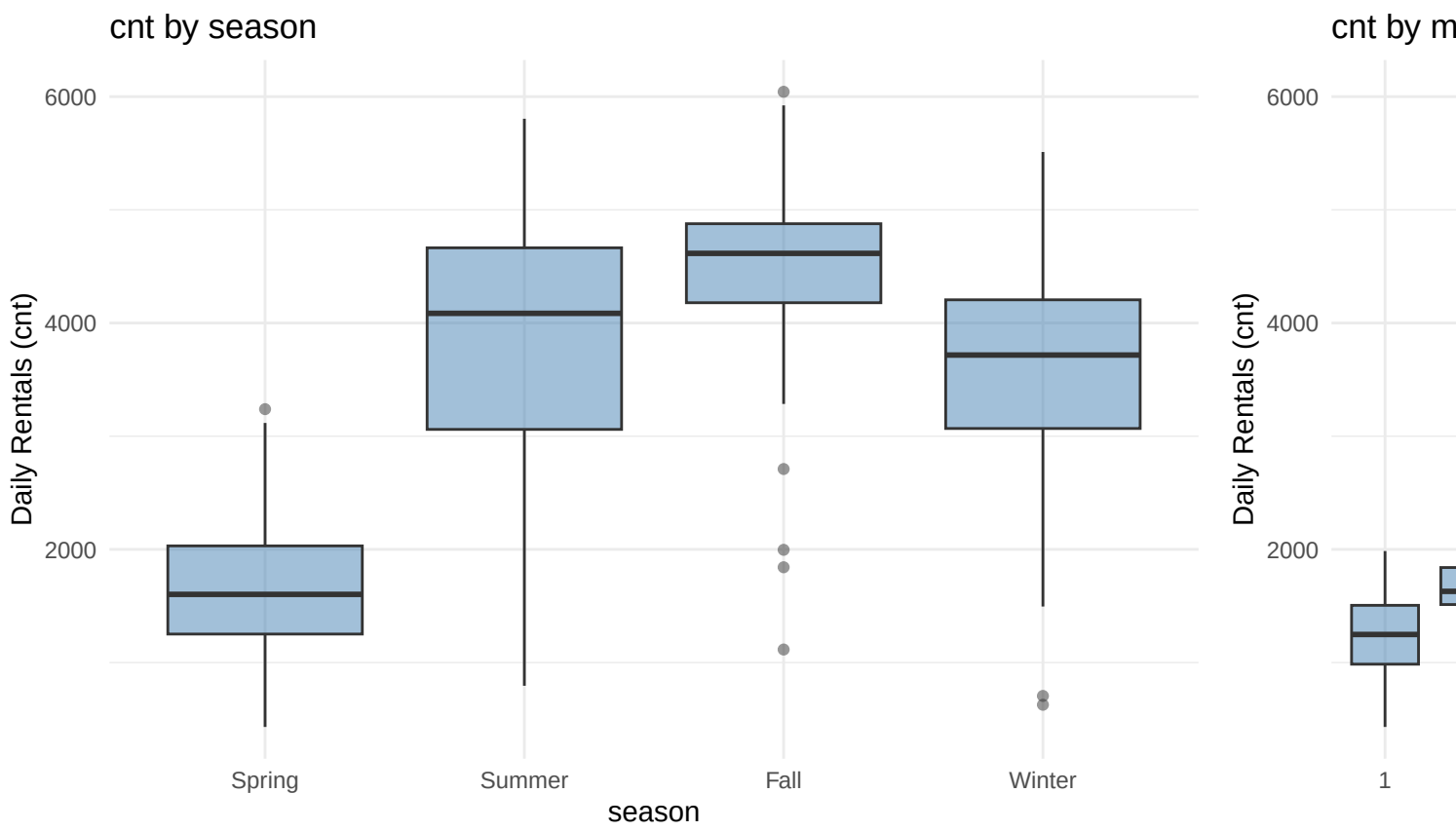


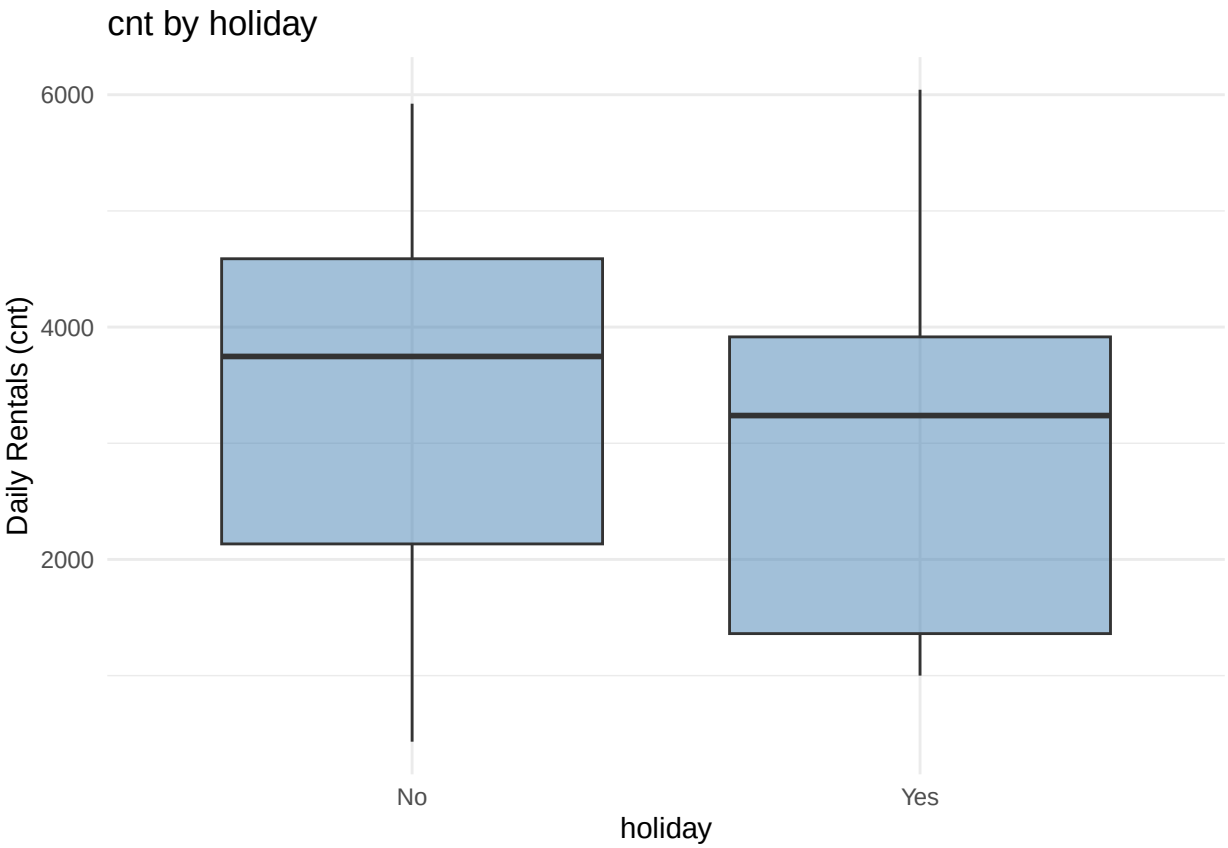


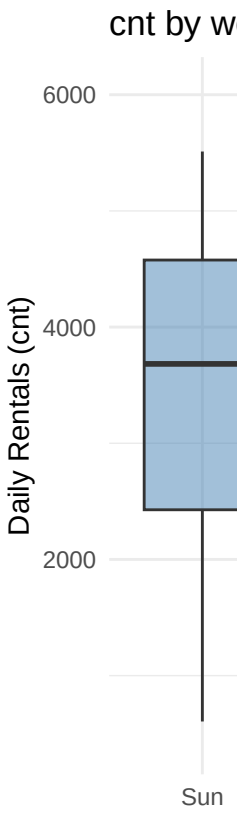
- cnt vs temp: there is a strong positive relationship
- cnt vs atemp: nearly identical to temp plot, so this confim multicollinearity
- cnt vs hum: almost a flat line indicating no relationship

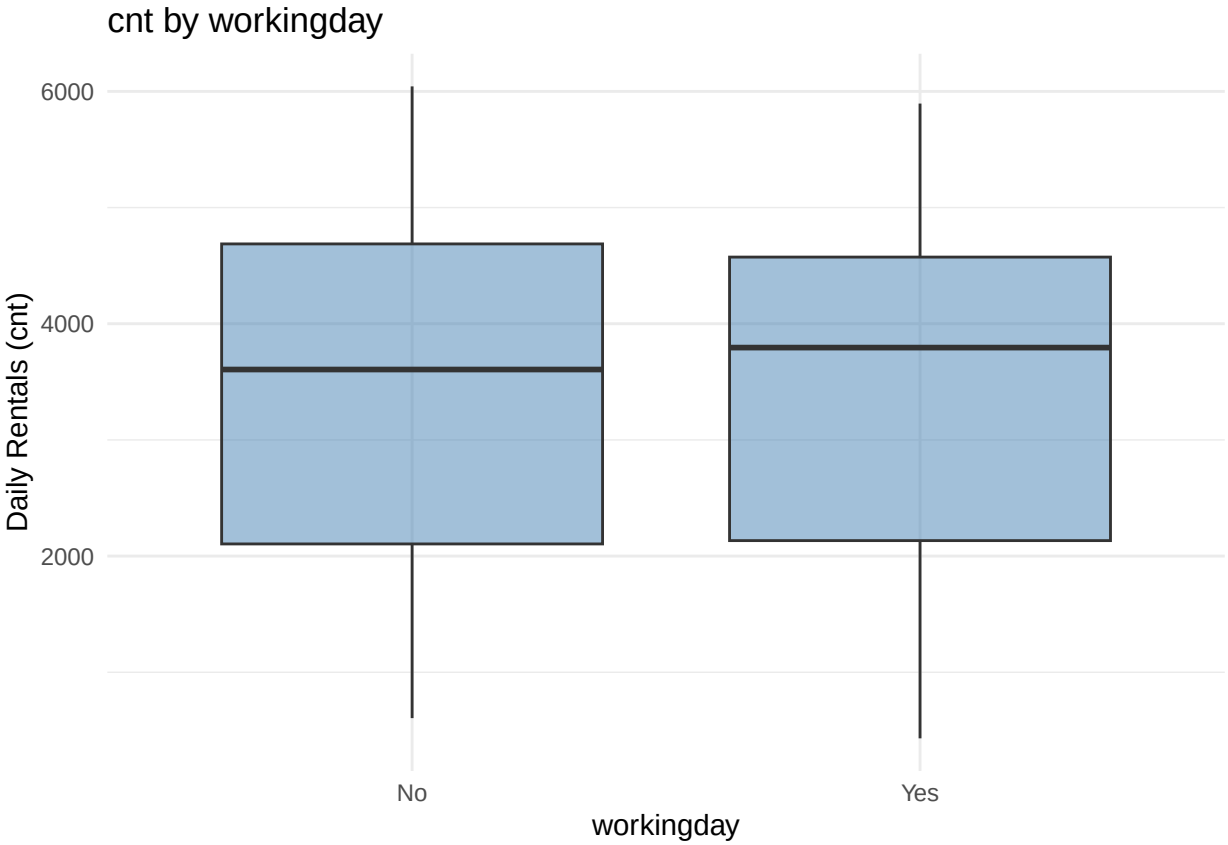
- cnt vs windspeed: there is a negative relationship

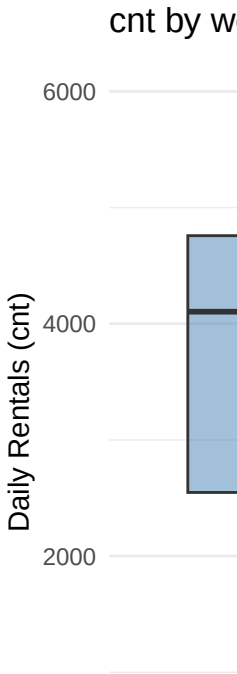
```
cat_preds <- c("season", "mnth", "holiday",
               "weekday", "workingday", "weathersit")

for(var in cat_preds) {
  p <- ggplot(bike_2011, aes(x = .data[[var]], y = cnt)) +
    geom_boxplot(fill = "steelblue", alpha = 0.5) +
    labs(title = paste("cnt by", var),
         x = var, y = "Daily Rentals (cnt)") +
    theme_minimal()
  print(p)
}
```











- There is a clear differences across seasons. Because of this differences, there will be variations model can expoloit therefore this going to be a strong predictor.

- cnt by mnth: There is an inverted U-shape. I would say that mnth captures the seasonal trend very granularly
- cnt by holiday: there is a slight differnce as holidays have slightly lower median rentals. the differnce is not significant. I would say this is a week predictor.
- cnt by weekday: boxes are very similar across all days. Hence, there is a little effect on rental. probably week predictor as well
- cnt by workingday: identical distributions. probably week predictor as well.
- weatherit: after we dropped the no observation for heavy rain/snow, there is a clear downward staircase pattern. That means as weather worsen so rental drop

Note that Light Rain/Snow box is very wide relative to its size, it would be unstable estimates due to only 15 observations

Summary:

```
##
## | Predictor      | Type         | Relationship with cnt                          |
## |----------------|--------------|------------------------------------------------|
## | temp / atemp   | Continuous   | Strong positive                                |
## | season         | Categorical  | Strong - clear group differences               |
## | mnth           | Categorical  | Strong - inverted U shape                      |
## | weathersit     | Categorical  | Strong - staircase drop                        |
## | windspeed      | Continuous   | Moderate negative                              |
## | hum            | Continuous   | Weak/flat                                      |
## | holiday        | Categorical  | Weak - overlapping boxes                       |
## | weekday        | Categorical  | Weak - nearly identical boxes                  |
## | workingday     | Categorical  | Weak - nearly identical boxes                  |
```